Frequency-Aware DETR Training at Extreme Class Imbalance

Technical report

Abstract

Logit adjustment is a standard remedy for long-tailed recognition, and applying it consistently across a DETR-family model's classification loss, Hungarian matching cost and denoising task is a natural extension. We report that at 34,320:1 imbalance it fails catastrophically: the unified treatment is 6.05 pp mAP below a stock baseline with tail AP at exactly zero, and loses 4.41 pp to conventional class-balanced reweighting. We identify the cause as magnitude rather than principle—at $\tau = 1.0$ the adjustment applies a $+11.47$ logit shift to the rarest class—and show recovery is monotonic in τ while never clearing the baseline. We further report a reproducible stability pattern: every arm applying the adjustment *inconsistently* diverged, while every consistent arm trained stably. Finally, decoupled classifier retraining is nominally $+0.53$ pp mAP on test, of which 98 % comes from a single class with *two* test instances.

1 Setup

Stock DINO-DETR [6], ResNet-50 4-scale, initialised from the official COCO checkpoint with a reinitialised 31-class embedding. $\mathrm{num\_classes} = 32$ because DINO indexes raw `category_id` without remapping.

All arms: 12 epochs (official $1\times$), `lr_drop` 11, batch 2, seed 42, identical initialisation and evaluation protocol. **No arm receives a hyperparameter search.** That keeps budgets matched; Section 4 shows it is also, in this instance, the defect.

Evaluated on a verified-disjoint split: 9752 / 2090 / 2090 images, 0 exact duplicates and 0 NCC-confirmed near-duplicates across 792,811 candidate pairs.

2 The Claim Under Test

In DETR-family models the matcher decides which query is supervised for which ground truth. If the classification loss is frequency-adjusted but the matching cost is not, matching keeps assigning tail ground truth to queries whose unadjusted scores look good while the loss optimises a different scale. The hypothesis is that applying the adjustment *consistently*—across loss, matching cost and denoising—beats applying it in one place, and beats ordinary loss reweighting.

The matrix was frozen before any arm ran. The control C1 applies per-class loss weights only [1]—no logit shift, nothing in the matcher or denoising—and was verified to reproduce stock sigmoid focal loss to **0.0 absolute difference** at unit weights, so the per-class weight is provably the only change.

3 Result

The unified treatment loses decisively to plain class-balanced reweighting, which in turn loses 1.64 pp to changing nothing. **Tail AP in every unified arm is exactly 0.0000**—the metric the method targets is the one it destroys. Layering oversampling [2] or contrast enhancement changes nothing, because the base configuration is already broken.

arm	configuration	mAP	AP50	AP75	mid	tail
D1	baseline	**0.1625**	0.3080	0.1551	**0.2118**	0.0368
D2	loss only			*diverged*		
D3	matching only			*diverged*		
D4	denoising only	0.1633	**0.3165**	**0.1572**	0.2029	**0.0455**
D5	**unified**	0.1020	0.1998	0.0939	0.0959	**0.0000**
D6	unified + oversampling	0.1014	0.1988	0.0946	0.0943	0.0000
D7	unified + contrast	0.1012	0.1993	0.0929	0.0972	0.0000
C1	plain reweighting	0.1461	0.2910	0.1375	0.1847	0.0270

Table 1: Validation, matched budget. The decisive comparison $D5 - C1$ is mAP -4.41, AP50 -9.12, AP75 -4.36, mid -8.87, tail -2.70 pp.

Only D4 clears the baseline, by 0.08 pp mAP. That is inside noise and carries no claim. On held-out test, D5 confirms: mAP 0.1001 against 0.1570 (-5.69 pp), tail 0.0000 against 0.0598.

4 Diagnosis: Magnitude, Not Principle

Logit adjustment [4] subtracts $\tau \log p(c)$ from each class logit. On this training set the rarest present class ($n = 1$) has log-prior -11.47 and the most common ($n = 34{,}318$) has -1.03: a spread of **10.44** in logit space. A $+11.47$ shift saturates the sigmoid for every query on that class.

Menon et al. validated $\tau = 1.0$ on CIFAR-LT and ImageNet-LT, whose imbalance is of order 100:1 to 1000:1. This corpus is **34,320:1**.

τ	mAP	Δ vs baseline	tail
1.00	0.1020	-6.05	0.0000
0.50	0.1340	-2.85	0.0135
0.25	0.1601	-0.24	0.0346

Table 2: Sweep on the unified configuration, changing nothing else. Recovery is monotonic in τ, confirming magnitude as the mechanism, and incomplete: even at $\tau = 0.25$ the method does not reach the baseline.

This is a transferability result about the published default. $\tau = 1.0$ is used across the long-tail literature without restatement of the imbalance regime it was tuned for. At clinical imbalance it is not merely suboptimal; it is destructive.

5 Divergence Is Patterned

Four arms failed with `inf` or `NaN` cost matrices at `linear_sum_assignment`. Which arms failed is not random: D2 (loss only), D3 (matching only), L1 (matching + denoising) and L2 (loss + denoising) all diverged, while D4 (denoising only) and D5–D7 (all three) were stable.

Checkpoints on disk are numerically clean—max $|w|$ 1.42e+01, identical to arms that completed—so divergence develops during the run. Gradient clipping was active at 0.1 throughout. When the loss carries a $+11.47$ shift and the matcher scores on unadjusted probabilities, the two components optimise against different scales and the assignment degenerates. **Consistency prevents divergence; it does not produce accuracy.**

We report this as an observation from crashes, not a designed stability experiment. The pattern is reproducible—D2 diverged on every attempt.

6 Decoupled Classifier Retraining

The long-tail literature indicates rebalancing belongs at the classifier stage [3, 5], not inside representation learning—the opposite of every arm above, and structurally unable to exhibit the failure of Section 5. We freeze all parameters except the classification embedding, start from the trained baseline, and retrain for 6 epochs under repeat-factor sampling.

On test this gives mAP 0.1623 against 0.1570 ($+0.53$ pp), AP50 $+0.88$ pp, tail $+1.25$ pp—positive on every metric, with the largest gain exactly where a long-tail method should deliver.

It does not survive decomposition. The largest single contributor is one class with **two test instances** moving $0.0000 \rightarrow 0.1515$, contributing $+0.52$ pp—**98 % of the total**. Excluding it, the gain is $+0.01$ pp. On the 18 classes with ≥ 10 test instances, 6 improve and 12 worsen.

7 Noise Floor (Measured)

The reference configuration was trained at three seeds, everything else identical. Determinism is exact on this pipeline, so seed choice is provably the only source of variation.

seed	mAP	AP50	AP75	head	tail
42	0.1055	0.2549	0.0661	0.2815	0.0101
1337	0.1056	0.2573	0.0679	0.2837	0.0117
2024	0.1037	0.2568	0.0629	0.2823	0.0105
sd	**0.0010**	0.0013	**0.0025**	0.0011	0.0008

Table 3: Noise floor at 2 sd: mAP ± 0.21, AP50 ± 0.26, AP75 ± 0.50, head ± 0.22, tail ± 0.17 pp.

Three consequences, and they are not the same verdict. The seven-arm ablation spans a 0.36 pp range, *inside* the band—selecting a winner on mAP was selecting noise. The best segmentation arm on test is $+0.21$ pp, sitting *exactly at* the threshold. But the isolated boundary effect on AP75 is $+0.70$ pp against a ± 0.50 pp floor, so it **exceeds** noise on validation at 50 epochs, and then reverses on test at 100 epochs. That is a **transfer failure, not a noise result**: the objective does what its mechanism predicts under the conditions it was measured in, and that does not survive a change of split and schedule.

8 Conclusion and Limitations

No configuration tested improves on stock DINO-DETR beyond noise. Consistent frequency-awareness loses 4.41 pp to plain reweighting, which loses to changing nothing; the pre-registered hypothesis is refuted by its own frozen matrix. The published $\tau = 1.0$ does not transfer to clinical imbalance, with failure monotonic in τ. Inconsistent application diverges reproducibly. Decoupled classifier retraining is the right mechanism and still does not clear noise. And group means over low-support classes fabricate results—twice, in two independent projects.

Single seed per arm, with the noise floor measured above. Single corpus. No external benchmark, and therefore no long-tail claim in the title, abstract or conclusions. Two cells are recorded as diverged rather than scored.

References

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